Author Response to Reviewer 1P2E

All changes marked in blue; line refs refer to revised `main.tex`.

R1.1: Theoretical discussion (bias-variance, parameterization complexity, dataset size). **Response:** Expanded simplicity principle discussion `[L604--605]`: explicit parameter-count analysis (∼180 vs. ∼7 samples/parameter for invariants vs. lowrank on ESOL), connected to model selection theory showing the bias-variance tradeoff penalizes higher-dimensional representations at these sample sizes (642–4,200 molecules). New feature dimension ablation (Section 5.6, `[L484--508]`; Table 6, `[L489--504]`): RMSE monotonically degrades D=256 → 2,048, confirming a fundamental statistical constraint. Conclusion updated `[L618]`.

R1.2: Classification benchmarks. **Response:** Added BACE (1,513 molecules, β-secretase inhibition) and BBBP (2,038 molecules, blood-brain barrier permeability): Section 5.5 `[L461--481]`, Table 7 `[L466--479]`. Conformer features hurt BACE (−4.4% AUROC) and show no benefit on BBBP (+0.1%)—consistent with the taxonomy, as β-secretase binding is an electronic/binding property and BBB permeability is governed by topological features rather than 3D geometry. Scope statement `[L151]` and abstract `[L80]` updated to include classification. Per-seed results in Appendix K `[L1041--1064]`.

R1.3: Larger/industrial datasets. **Response:** Acknowledged as important limitation `[L620]`. Benchmarks span 642–133K molecules (MoleculeNet, QM9, MARCEL); validation on ChEMBL/ZINC/proprietary ADMET panels noted as priority for future work.

R1.4: Figure resolution, standardize metrics, typos. **Response:** All figures re-exported at ≥300 DPI. Evaluation metrics standardized: RMSE throughout for regression, AUROC for classification `[L461--481]`. Full proofreading pass completed; typographical errors and formatting inconsistencies corrected across all tables and main text (see blue-marked changes throughout).

R1.5: Scaffold split generation and replicates. **Response:** Expanded Training Details `[L274]`: Bemis–Murcko generic scaffolds extracted via RDKit, sorted by frequency, assigned to train/validation/test maintaining 80/10/10 ratios. Seed counts made explicit: 3 seeds for main benchmark, 10 seeds for statistical validation (sufficient power to detect ∼5% effect sizes at α = 0.05). Classification benchmarks use stratified random splits `[L464]` since scaffold splits on imbalanced datasets yield one-class test sets.